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Design and allocation of power system stabilizers using the particle swarm optimization technique for an interconnected power system

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ABSTRACT

In this paper, three particle swarm optimization (PSO) based power system stabilizers (PSSs) are developed for three power systems. The system under study here is a power pool consisting of 3 power systems. System I represents the Egyptian power system, system II represents the Jordan and Syrian power systems, and system III for the Libyan power system, which are originally self standing and completely independent systems. As a matter of fact each of them should equipped with its own PSS. For this reason this work is started by designing an optimum power stabilizer for each of them standing alone. After which, the developed PSSs are firstly installed one at a time. Then the three PSSs are installed together in the interconnected power system and their effect on its dynamic performance is studied.

As a test for stabilization efficiency, the detailed power system model is subjected to a forced outage of a 600-MW generator, which is the biggest unit in the pool, when it is fully loaded. This outage results in loosing of about 3% of the spinning capacity of system I and about 2% of the spinning capacity of the whole interconnected system. The obtained results show an improvement in the power pool performance accompanied with an improvement in the inter-area oscillation.

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1. Introduction

Power systems are complex non-linear systems and often exhibit low-frequency electromechanical oscillations due to insufficient damping caused by adverse operating conditions. Low frequency oscillations can severely restrict system operation and can reduce power system security level. Power system stabilizer (PSS) units have long been regarded as an effective way to enhance the damping of electromechanical oscillations in power system [1].

Some research works concerning the use of PSSs in practical power systems are published in [2–4]. In [2], the authors investigate the coordinated use and tuning of PSSs to improve small-signal dynamic performance of the Mexican interconnected system. In [3], allocation of PSSs designed by a frequency response method in an interconnected system has been determined by an eigenvalue analysis. Ming et al. [4] describes a comprehensive small-signal analysis for China Southern Power Grid in 2005 and explores the optimal control strategies for both PSS control and HVDC stabilizers by means of coordinated parameter optimization.

Many artificial intelligence techniques have been tested to achieve tuning under various operating conditions for hypothetical power systems [5–9]. Chiou et al. [5] developed a mixed-integer ant direction hybrid differential evolution algorithm for tuning the PSS for a multimachine power system. A comparison between a small population based particle swarm optimization (SPPSO) and bacterial foraging algorithm (BFA) for simultaneous tuning of a number of PSSs in two area with four machine power system is presented in [6]. El-Zonkoly et al. [7] uses the PSO to optimize a lead-lag and Fuzzy logic PSSs parameters while the PSO technique is used in [8] to optimize the PSS and AVR gains for each generator of the New England 39-bus system.

A laboratory model is belt and the results of real time implementation are proved for three neuro-fuzzy PSSs; each is tuned using genetic algorithm [9]. Bevrani et al. [10] performs a real-time experiment for longitudinal four machine infinite-bus system to illustrate a new robust PID-based PSS to find an optimal gain vector via an H_∞ static feedback control.

In order to overcome the problem of interactions between stabilizers, coordinated design of multiple controllers has been proposed [11–13]. The application of a decentralized modal control method for pole placement in multimachine power systems utilizing FACTS (Flexible AC Transmission Systems), STATCOM (Static Synchronous Compensator) and UPFC (Unified Power Flow Controller) devices are presented in [11]. Comparative analysis on

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the function of damping of the FACTS, STATCOM and UPFC is performed using the New England 39-bus system.

A design procedure for simultaneous coordinated designing of the thyristor controlled series capacitor (TCSC) damping controller and power system stabilizer (PSS) in multi-machine is developed in [12]. The coordinated design problem is solved using PSO.

Three examples of multi-machine power systems installed with a UPFC, a STATCOM/BESS (Battery Energy Storage System) and a MUPFC (Multiple-terminal UPFC) respectively are presented in [13]. Parameters of multiple stabilizers are designed in coordination by using a newly appeared method of optimization – artificial fish swarm algorithm.

In a previous work of the authors, a reduced system model for the power pool consisting of three interconnected power systems has been developed [14]. The basic feature of the developed model is that its parameters are intentionally adjusted to meet the characteristics of an actual interconnected power system.

The system model developed in [14] is used here to examine a proportional-derivative (PD) PSS tuned by the PSO technique. For simplicity, the PSSs are designed for three simplified systems that are considered equivalent to the actual power systems of the power pool. Each simplified power system consists of a single machine-infinite bus system (SMIB) [15]. All designs are made for one operating condition of the system.

The obtained PSSs are tested firstly on the equivalent SMIB systems in which they are designed. Then, the control gains found for the three PSO-based PSSs are fed into the overall model of the power pool including the three interconnected power systems. This model has been built in the MATLAB-SIMULINK [16] for the power pool.

2. The study system

The interconnected power system consists of six equivalent generators (G_1 for all the thermal power stations in Egypt except G_2, G_3 for High Dam hydro station, and G_4 for the Aswan Dam hydro stations 1, and 2, G_5 equivalent to the Jordan and Syrian power systems, and G_6 equivalent to Libyan power system), 14 equivalent loads and 3 HV and EHV long transmission systems as shown in Fig. 1.

System I consists of two powerful subsystems I_a and I_b which are interconnected by a long 500-kV, double-circuit, 4-section strong over head transmission line (OHTL) in parallel with another 220-kV, double-circuit 4-section OHTL. This system includes more than 20,000 MW installed generation capacity and has a peak load of about 18,000 MW. The generation capacity and the load are distributed among the two subsystems I_a and I_b [17]. The transmission systems interconnecting subsystems I_a and I_b are represented in detail to simulate reactive power sources and intermediate loads connected to them and to have a focus on the effect of reactive power sources on the overall system performance.

The parameters of generating unit G_5, together with its control loops are set at normal values to get a reasonable realistic tie line loading in case of disturbances and different events.

The reactive power requirements of the 500-kV single-circuit tie line between systems I and II is supplied by both systems.

The tie line between systems I and III consists of three equal 220-kV double-circuit sections, 200 km long each, connected in series. The line is equipped with two inductive reactive power compensators, 50 MVAr each, connected at one third and two thirds of the line length. The purpose of such reactors is to compensate for the effect of the shunt capacitance of the line.

The parameters of the generating unit G_6 together with its control loops (AVR and LFC) are set equal to their normal values to get a realistic tie line loading in case of disturbances.

The tie line of G_5 with system I is a 500 kV single-circuit line and is relatively stronger than the 220 kV double-circuit line connecting system III with system I. The data of the equivalent generators and loads are shown in Tables 3 and 4 (Appendix A).

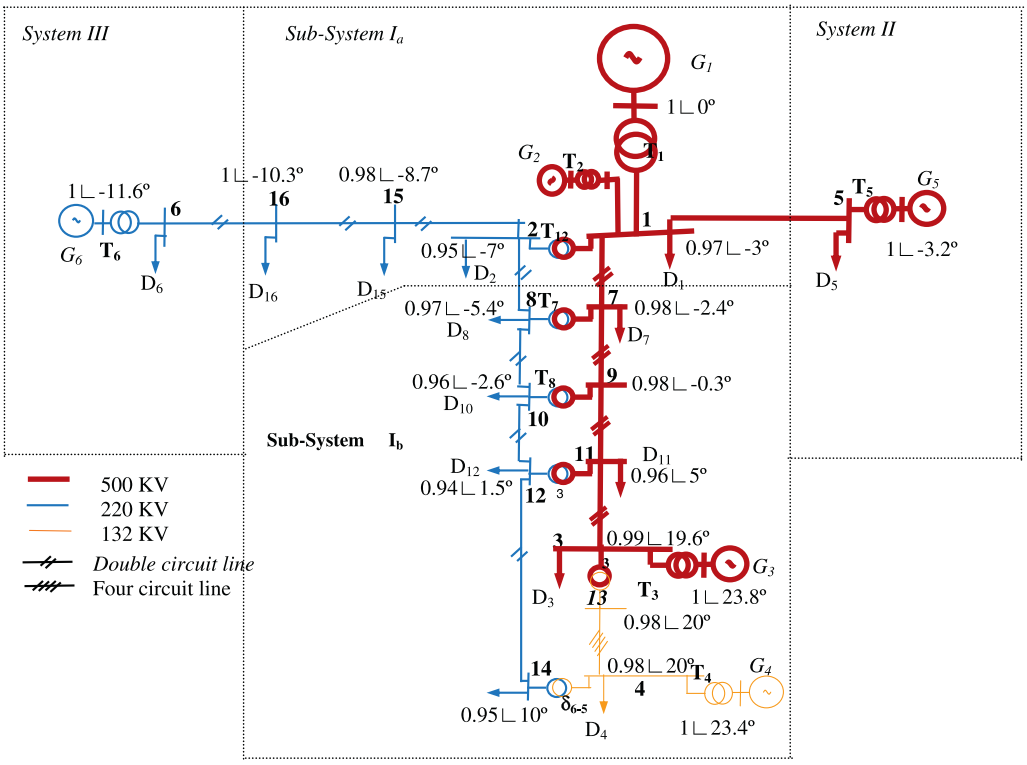


Fig. 1. Single-line diagram of the study power system.

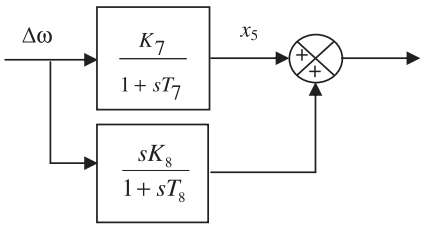


Fig. 2. The PD PSS model.

3. Designing a PSS using the PSO technique

The 500 kV busbar number 1 in system I (as shown in Fig. 1) is considered as an infinite bus for systems II and III when designing the PSSs to be installed on G_5 and G_6 and busbar number 11 is considered as an infinite bus when designing the PS for generator G_3.

A PD PSS type with $\Delta\omega$ as input signal is chosen. The block diagram representing this stabilizer is shown in Fig. 2.

Governed by practical consideration, the time constants T_7 and T_8 are taken equal to 0.05 s, where K_7 and K_8 to be tuned by the PSO technique. The optimization process is based on one of the following two approaches:

1. Shifting system roots in the complex root plane as much as could be possible.
2. Maximizing the least damping ratio of the stabilized system eigen modes.

Both approaches are tested and it is found that the first approach is more effective in tuning the stabilizer.

A particle swarm optimization toolbox (PSOt) for use with the Matlab scientific programming environment has been developed in [18]. It is modified and employed to get the optimal values of parameters according to the given predefined ranges. In addition to the main PSO program, additional programs were designed to get the state-space representation of the system.

3.1. Objective function

To increase the system damping, the eigenvalue-based objective function is considered as follows:

$$J = \max\{\mathrm{Real}(\lambda_i)\} \tag{1}$$

where λ_i is the ith electromechanical mode eigenvalue. In the optimization process, it is aimed to minimize J in order to shift the poorly damped eigenvalues to the left in S-plane. The optimization problem can be formulated as follows:

$$\text{minimize}(J)\text{subject to} \tag{2}$$

$$\begin{aligned} k_7^{\min} \leqslant k_7 \leqslant k_7^{\max} \\ k_8^{\min} \leqslant k_8 \leqslant k_8^{\max} \end{aligned} \tag{3}$$

Typical ranges of the optimized parameters are [−500: 500] for all gains. The results obtained by the PSO technique for the PSSs control gains for the three generators G_3, G_5 and G_6 are shown in Table 1.

Table 1
PSO-tuned PSSs for G_3, G_5, and G_6.

Generator	K_7	K_8
G_3	68.24	318.3
G_5	5.6	−2.84
G_6	4.47	−3.87

4. Simulation results

4.1. Simplified power systems

The PSSs designed in the last section are tested firstly on the single machine-infinite busbar systems for which they are designed. The input disturbance assumed is $\Delta Tm = 0.01$pu. The deviations of the total power angle and its rate of change are recorded for each of the generators G_3, G_5 and G_6 for which the PSSs are designed by the PSO technique [19].

4.2. Interconnected power system

The control gains found in the previous section for the three PSO-tuned PSSs are fed into the power pool model that has been built in the MATLAB-SIMULINK. The excitation system and the automatic voltage regulator (AVR) are represented by the MATLAB-SIMULINK model given in [15]. This model is simplified and combined with the third order machine model given in [13] and the PSS is designed for the combination.

The PSS application alternatives are:

(i) The generators G_3, G_5, and G_6 are equipped, one at a time, with their PSSs.
(ii) The three PSSs are simultaneously equipped with their generators.

The selected operating point is somewhat below the peak loading of the system and the voltage levels are set somewhat below their maximum limits. These voltage levels are selected intentionally to worsen the system stability problem.

The initial conditions corresponding to the operating point under consideration are set by the load flow utility of the POWERGUI of the MATLAB-SIMULINK and automatically sent to the blocks representing different items.

As a test for stabilization efficiency, the power pool is subjected to a forced outage of the 600-MW generator G_2 when it is fully loaded. This outage results in loosing of about 3% of the spinning capacity of system I and about 2% of the spinning capacity of the whole interconnected system.

The system response to the forced outage of the fully loaded generator G_2 has been found for a time period of 2 min directly after disturbance occurrence. Such two minutes period is sufficient to judge the effectiveness of stabilization. However, in some cases the recording of the disturbed system response continued for 10 min after the application of the disturbance. The relatively long period is taken to make sure of system convergence.

The most important meaningful and indicative system variables inspected are:

- System frequency.
- Power angle of generators G_3, G_5, and G_6 with respect to the slack bus.
- Active and reactive powers outputs of G_3, G_5, and G_6.
- Tie line powers between system I and systems II and III.
- Relative power angle of G_5 with respect to G_6.

These variables are recorded and plotted in the time domain, and the most important of them is illustrated below to judge the dynamic performance of the system and the effectiveness of the PSSs being tested.

4.2.1. Case 1: PSO-tuned PSS acting on generator G_3

The system response obtained in this case is demonstrated in Fig. 3a–g. The response of the original system (i.e. system without

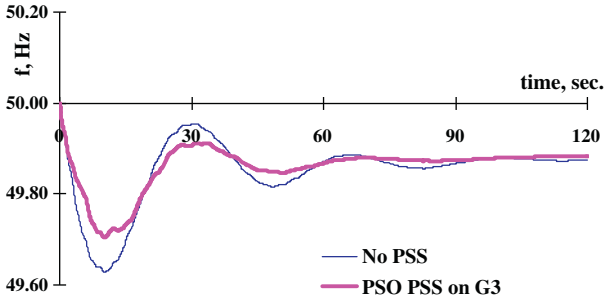


Fig. 3a. System frequency for PSO-based PSS acting on G_3.

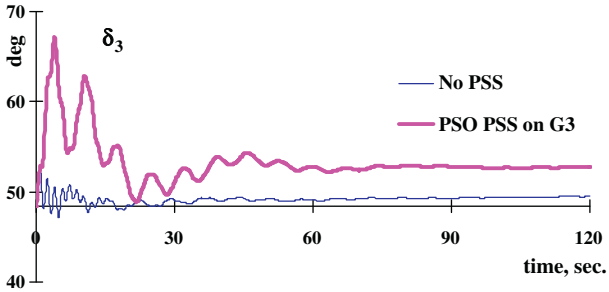


Fig. 3b. G_3 power angle.

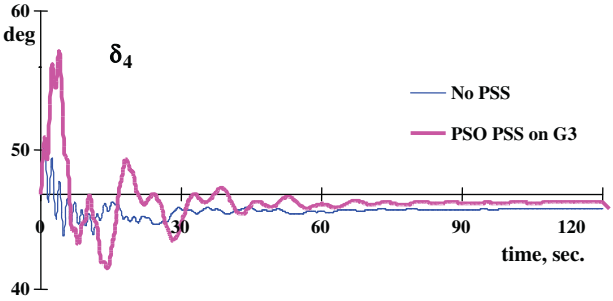


Fig. 3c. G_4 power angle.

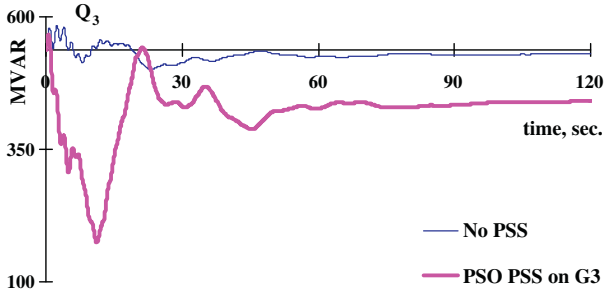


Fig. 3d. Reactive power output of G_3.

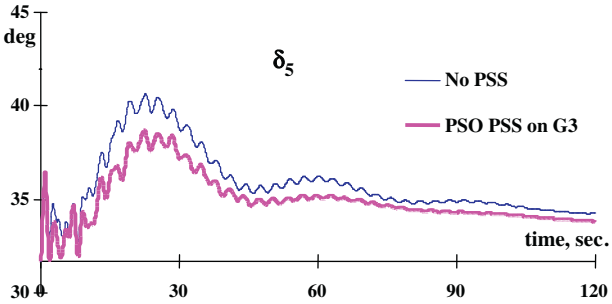


Fig. 3e. Power angle of G_5.

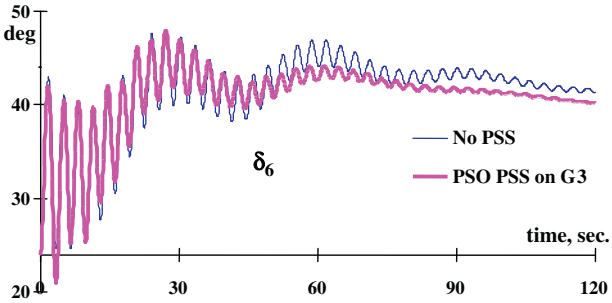


Fig. 3f. Power angle of G_6.

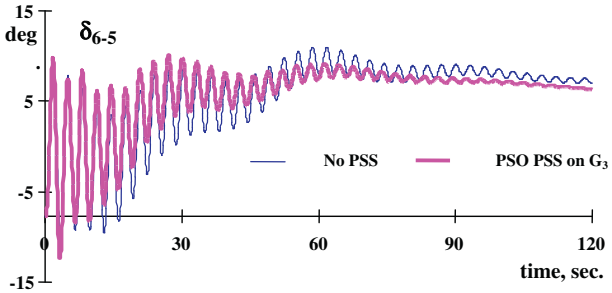


Fig. 3g. Power angle of G_6 with respect to the q-axis of G_5.

the PSO–PSS) to the same disturbance is also shown on the same figure for comparison purpose.

Fig. 3a shows that system frequency response has been significantly improved after applying the PSO-tuned PSS on G_3. The system frequency reached a minimum value of 49.704 Hz while it is 49.628 Hz without using the PSS. It could be also remarked that better damping of frequency oscillations has been achieved.

Figs. 3b and 3c show that more oscillations have been recorded in rotor power angles for G_3 and G_4 respectively. Similar oscillations are found in the reactive power output of G_3 as shown in Fig. 3d. On the other hand, these oscillations are associated with an increased share of G_3 in the total system power to compensate for the generation deficiency due to the outage of G_2.

Figs. 3e and 3f show that the improvement in system frequency response due to the application of the new PSS on generator G_3 is associated with additional improvement in the power angle oscillations of both G_5 and G_6.

Fig. 3g indicates that mutual oscillations between G_5 and G_6 are became more damped.

It could be concluded that the installation of a properly tuned PSS on G_3 could result in improving the overall system frequency and also the inter-area oscillations of systems II and III with respect to system I and between them.

4.2.2. Case 2: PSO-tuned PSS acting on generator G_5

System response to the forced outage of the fully loaded generator G_2 has been found for this case and the obtained results are demonstrated in Figs. 4a–e. The obtained response is compared with that of the system without additional stabilization in the same figures.

Fig. 4a shows that the installation of a properly tuned PD PSS on G_5 has resulted in a little-bit improvement in system frequency. In this case, the minimum frequency reached 49.662 Hz which is lower than that obtained with the same type of stabilizer on G_3 (Fig. 3a).

Figs. 4b and 4c show that there is a slight improvement in system response which is a result of improved oscillations of G_5 and, consequently, the active power injected from system II to system I.

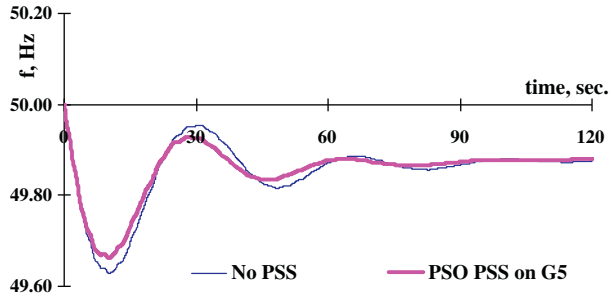


Fig. 4a. System frequency for PSO-tuned PD PSS acting on G_5.

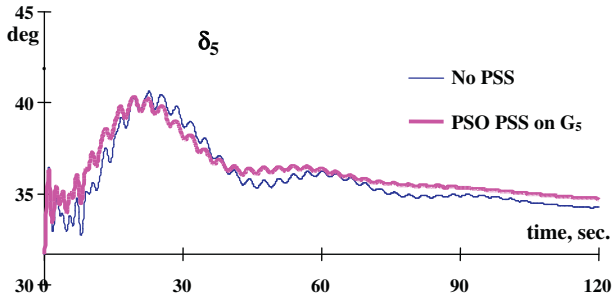


Fig. 4b. Power angle of G_5.

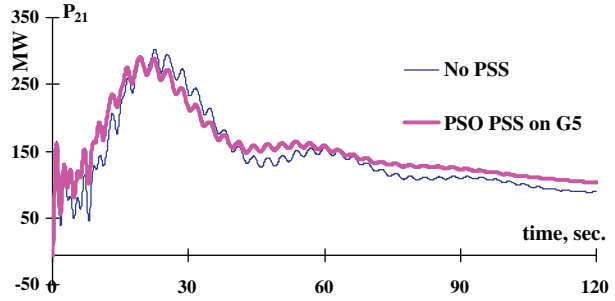


Fig. 4c. Active power injected from system II to system I.

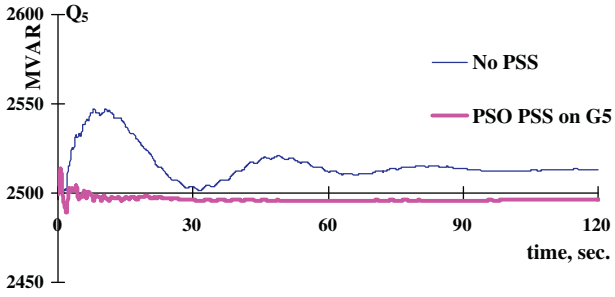


Fig. 4d. Reactive power output of G_5.

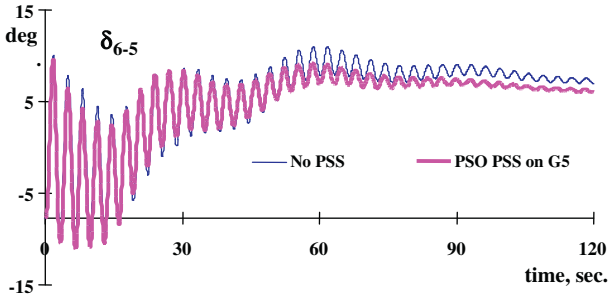


Fig. 4e. Power angle of G_6 with respect to the q-axis of G_5.

The contribution of G_5 in improving system performance is mainly due to its share in compensating for power deficiency in system I as shown in Fig. 4d.

Mutual oscillations between G_5 and G_6 are also slightly improved as shown in Fig. 4e.

4.2.3. Case 3: PSO-tuned PSS acting on generator G_6

In this case, the system is subjected to the same disturbance as in cases 1 and 2 above (forced outage of G_2). The corresponding system response has been found and illustrated in Figs. 5a–e.

Fig. 5a shows that a slight improvement in system frequency has resulted after installing the PSO-tuned PSS on G_6. The minimum frequency recorded is 49.644 Hz which is a slightly better (0.016 Hz) than in the case without the PSS.

The slight improvement in system response is due to slight improvement in injected active power from system III to system I due to more sensible stabilized power angle oscillation of G_6 with respect to system I as shown in Fig. 5b.

The main reason for the improvement in system response in this case is the direct effect of the new PSS on the reactive power output of G_6 as shown in Fig. 5c.

Fig. 5d shows a reasonable improvement in power angle of G_5 as compared to the weak contribution of the PSS acting on the far generator G_6.

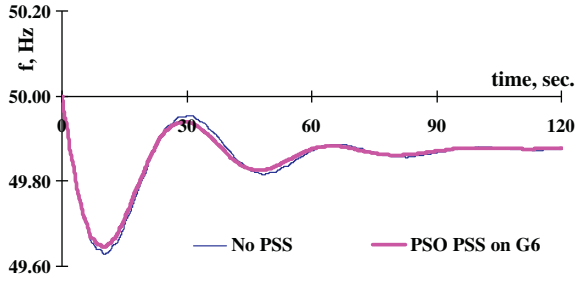


Fig. 5a. System frequency when a PSO-tuned PSS is acting on G_6.

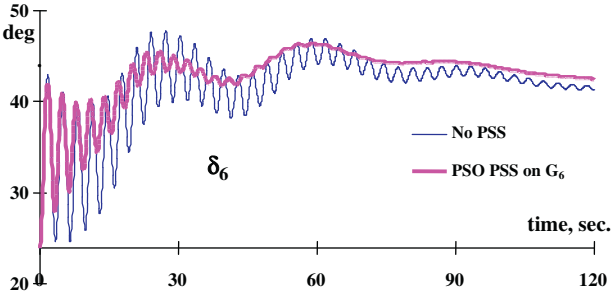


Fig. 5b. Power angle of G_6.

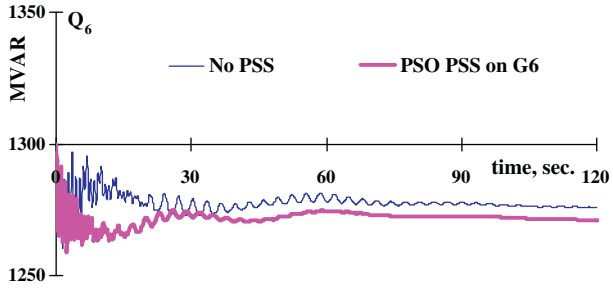


Fig. 5c. Reactive power output of G_6.

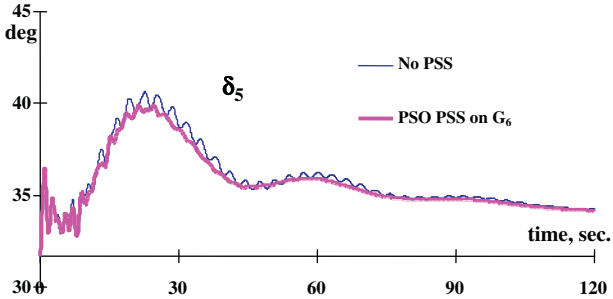


Fig. 5d. Power angle of G_5.

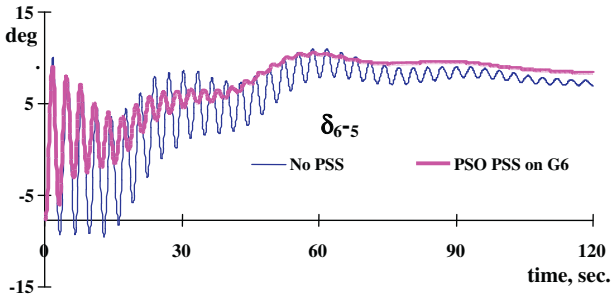


Fig. 5e. Power angle of G_6 with respect to the q-axis of G_5.

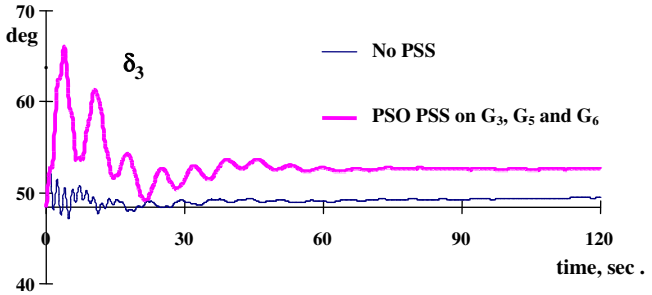


Fig. 6b. Power angle of G_3.

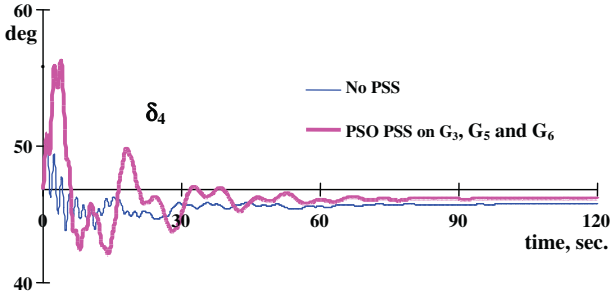


Fig. 6c. Power angle of G_4.

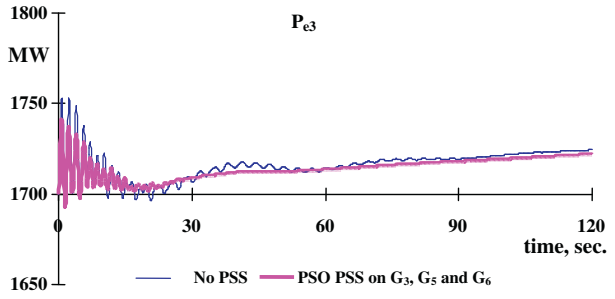


Fig. 6d. Active power injected from G_3.

It is logical to have the inter-area oscillations between system II and system III significantly improved as shown in Fig. 5e for the power angle between G_5 and G_6.

4.2.4. Case 4: three PSO-tuned PSSs acting simultaneously on G_3, G_5 and G_6

In this case, we are interested in studying the combined effect of the three PSSs if they work simultaneously on the overall system performance. The obtained results are demonstrated in Figs. 6a–6l. Fig. 6a shows how system frequency is affected by the forced outage of G_2 in the presence of the PSO-tuned PSSs on G_3, G_5, and G_6. Comparing this figure with Figs. 3a, 4a, and 5a, it can be easily remarked that the system frequency dip in this case (0.269 Hz) is the smallest one. However, it is not much better than that obtained with one PSO-tuned PSS installed only on G_3.

Figs. 6b and 6c show the power angles of G_3 and G_4, respectively, in this case. The point of main concern here is the role

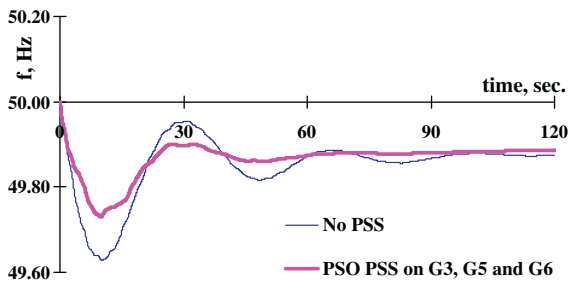


Fig. 6a. System frequency with PSO-tuned PSSs acting on G_3, G_5 and G_6.

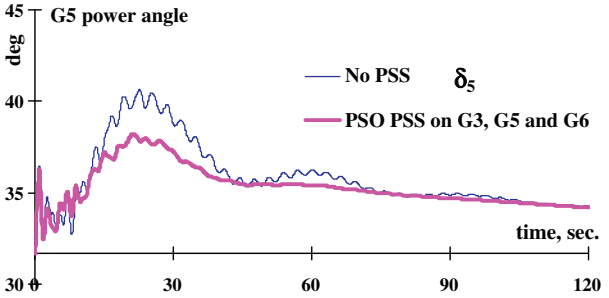


Fig. 6e. Power angle of G_5.

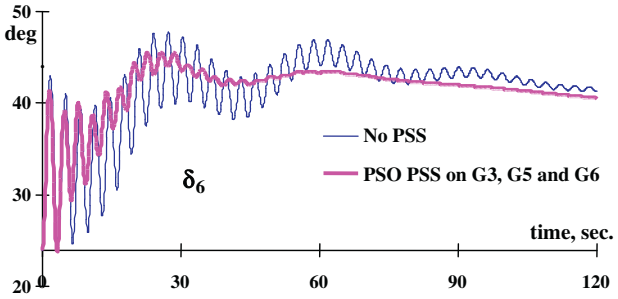


Fig. 6f. Power angle of G_6.

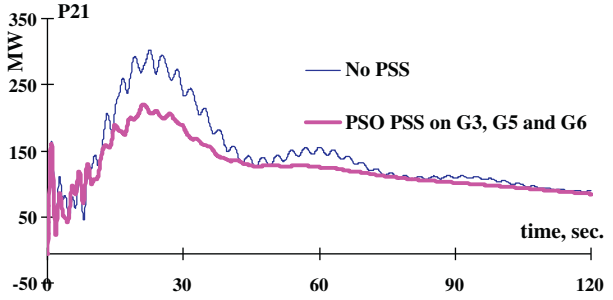


Fig. 6g. Active power injected from system II to system I.

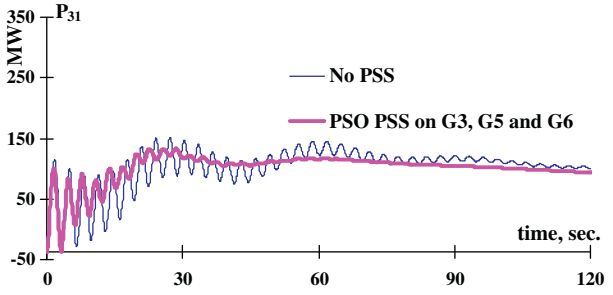


Fig. 6h. Active power injected from system III to system I.

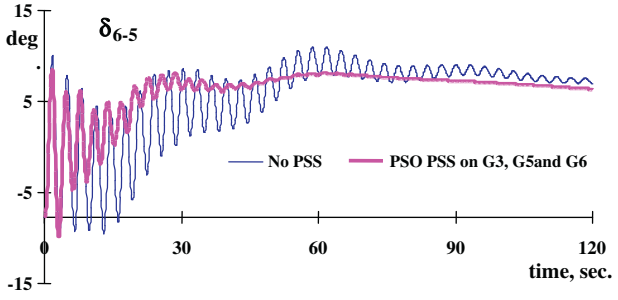


Fig. 6i. Power angle of G_6 with respect to the q-axis of G_5.

played by the active power injected from G_3 into the system during the first 15 s following the disturbance in improving the total system response as shown in Fig. 6d.

Figs. 6e and 6f show a remarkable improvement in the oscillatory response of the power angles of G_5 and G_6 in the presence of PSO-tuned PSSs on the three generators.

The response of active power injected from system II (G_5 side) and system III (G_6 side) to the central system I is improved as shown in shown in Figs. 6g and 6h.

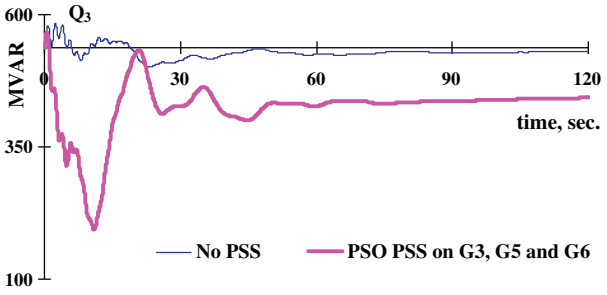


Fig. 6j. Reactive power output of G_3.

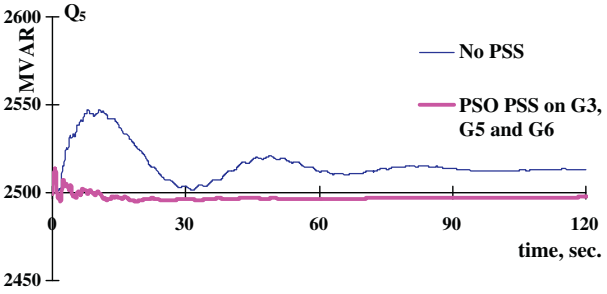


Fig. 6k. Reactive power output of G_5.

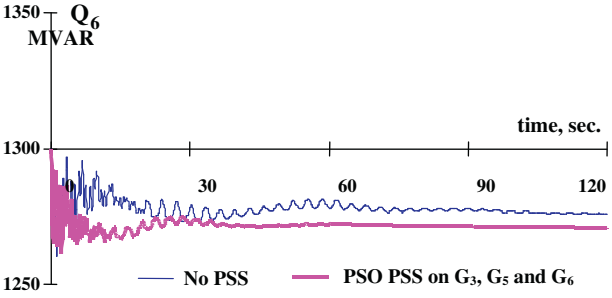


Fig. 6l. Reactive power output of G_6.

Fortunately, it should be also noticed that the inter-area oscillations between system II and system III (mutual power angle between G_5 and G_6) have been improved as shown in Fig. 6i.

Figs. 6j, 6k, and 6l show, respectively, a comparison of the reactive power output of G_3, G_5, and G_6 and the corresponding quantities in the case where no PSO-tuned PSSs are used.

5. Comments on simulation results

The system frequency is taken as a measure for the evaluation of improvements in the interconnected system performance when PSO-tuned PSSs designed for single machine infinite bus systems are used in the interconnected power system.

Applying the new PSSs one at a time or simultaneously on the three generators G_3, G_5, and G_6 has resulted in an additional improvement in the overall system performance.

The most effective location for applying only one PSS in the interconnected system is at G_3.

Using a PSO-tuned PSS on G_5 gives good results but not as good as that installed on G_3.

A slight improvement in system performance takes place when a PSO-tuned PSS is installed on G_6.

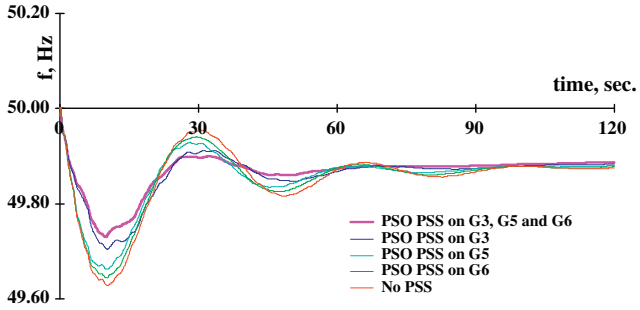


Fig. 7. System frequencies for different alternatives of PSO-tuned PSSs.

Table 2
Effect of different locations and compinations of PSSs on maximum DIP in system frequency due to a forced outage of a 600 MW generating unit.

Case order	PSO-tuned PSS acting on generator (s)	Maximum frequency dip (Hz)
0		0.372
1	G_6	0.356
2	G_5	0.338
3	G_3	0.296
4	G_3, G_5 and G_6	0.269

The best performance is obtained when G_3, G_5, and G_6 are simultaneously equipped with the three PSSs. The frequency dip in this case is the least one as it dropped to 0.269 Hz. The system frequency for the four cases as compared with the case without any PSS is shown in Fig. 7.

All studied combinations of the PSS PSO-tuned PD PSS installed on the three generators G_3, G_5, and G_6 are arranged in Table 2 in a descending order according to the magnitude of the frequency dip following system disturbance.

From generator power angle oscillations point of view, using PSO-tuned PSS on either G_5 or G_6 improves their own oscillations. The contribution to total system performance improvement comes from improving the performance of the two equivalent generators G_5, and G_6. This improvement has found a reflection on the interchanged active power between system I and systems II and III (according to the response of power angle of G_5 or G_6) and reactive power (according to excitation and terminal voltage of G_5 or G_6 which are found but not shown here).

The difference in system response when equipped with one PSS on G_3 from its response when equipped with the same type of stabilizers on G_5 or G_6 one at a time may be due to the following reasons:

- G_3 is hydro-alternator while both G_5 and G_6 are thermal units.
- The different nature of transmission systems connecting G_3, G_5, and G_6 with the equivalent huge generator G_1. While G_3 is connected with G_1 via a strong transmission system, G_5 and G_6 are connected with G_1 via a relatively weak tie lines.
- The loading conditions of the simple systems in which the PSSs are tuned using the PSO technique.

According to the previously described differences, the contribution of the PSO-tuned PSS of G_3 is mainly to improve the total system performance by increasing the participation of G_3 in facing the deficiency in power due to the outage of G_2. On the other hand, the PSS of G_5 or G_6 improved the overall system performance through stabilizing the response of the generators themselves, which is reflected on the total system performance.

Another difference is remarked between the effectiveness of PSO-tuned PSSs installed on G_5 and G_6 that the improvement in system performance contributed by these two generators is not in proportion to their power capacity. As stated before, only a slight improvement has been remarked by adding a PSS on G_6. The main reason comes from the difference in the nature of the tie lines of both systems.

The obtained results show that any improvement in system response is also accompanied with an improvement in the inter-area oscillations between systems II and III from one side and system I in the other side.

One main common remark is the decrease in terminal voltage of the generator on which the PSS is installed (within accepted operating limits). This reduction of voltage acts to reduce power consumption of the load connected to generator terminals and, thus, takes part in facing the power deficiency situation.

Another remark to be taken into account is that the frequency recovery after the power deficiency disturbance acts to slow-down the rate of rise of mechanical input to all generators running in the system. This result is due to the reduction in frequency deviation from nominal value (input signal to the primary LFC) and reduction in the integral value of frequency deviation from nominal (input signal to the secondary LFC).

Finally comparisons indicated that; although the PSSs are designed for simplified single-machine infinite busbar models, they are, fortunately, still efficient in improving the complete system dynamic response. Also no serious negative mutual interactions have been detected between different PSSs. This may be attributed to the fact that the PSSs are located far from each other, which is a specialty for the study system.

6. Conclusion

The system under study here is a power pool consisting of three practical power systems. The systems I, II, and III are originally self standing and completely independent systems. As a matter of fact, each of them is equipped with its own power system stabilizer. For this reason the work demonstrated in this paper is started by designing an optimum power stabilizer for each of these systems using the PSO technique. After that, the developed PSSs are installed; (i) one at a time, and (ii) concurrently on the three generators of the interconnected power system, and their effect on the dynamic performance of the system has been studied. The obtained results ensure that the most effective stabilizer is that installed on G_3 in system I.

The main contribution of this work is that, instead of installing only one PSS in one of the interconnected systems, which is the usual case, three PSSs are installed one in each of them without serious negative mutual interactions between the different PSSs. This may be attributed to the fact that the PSSs are located far from each other, which is a specialty for the study system.

Thanks to this situation, each individual system will have its own efficient PSS incase the power pool is subjected to normal or abnormal operating condition leading to separating them from each other.

Appendix A

See Tables 3 and 4.

Table 3
Ratings of the equivalent generators.

	G_1	G_2	G_3	G_4	G_5	G_6
Rotor type	Round rotor		Salient pole		Round rotor	
	Thermal	Mach.	Hydro	Mach.	Thermal	Mach.
Rated power (MW)	16,000	600	2100	595	7500	4000
Nominal voltage (kV)	20		15.75	11	20	
Frequency (Hz)	50					

Table 4
Load data.

Load	MW	Type	MVAR	Type
D1	12,550	1	6360	2
	627	2	310	2
D2	720	1	400	2
D3	1	2	330	2
D4	150	2	75	2
D5	5100	2	2600	2
D6	2800	2	1300	2
D7	200	2	100	2
D8	170	2	85	2
D14	200	2	100	2
D12	520	2	280	2
D10	330	2	175	2
D16	1	2	50	2
D15	1	2	50	2
D11	565	1	285	2

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